TM Forum Specification

Outage Management User Guide

TMF777

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Introduction

This document is the user-guide of the REST API for Outage Management. It includes the data model definition as well as all available operations for the Outage resource.

The Outage Management API provides a standardized mechanism for Outage management such as the creation, update, retrieval, import and export of a collection of outages. An Outage represents a record of a formally declared period of disruption or interruption of service(s) in a specific geographical region(s), preventing users from effectively accessing or utilizing the service. This could be planned or unplanned. It is comprised of attributes and associations which define all aspects known for a particular type of Outage.

The Outage management API will provide comprehensive outage details, including impacted services, regions affected, and the duration of the outage with estimated restoration times, allowing telecom operators to efficiently handle service disruptions for customers.

The Outage Management API manages the Outage resource:

- An Outage is comprised of attributes and associations which define all aspects known for a particular type of Outage.
- The main attributes are its impact, isPlanned, reason, state, estimatedDateToResolve, affectedService, affectedLocation and related incident or change request.
- An Outage follows a simple lifecycle which is presented later for illustrative purposes, defining such aspects as whether it is declared, restored or closed.

The Outage Management API performs the following operations on the Outage resource:

- Retrieval of a Outage or a collection of Outage, depending on filter criteria
- Partial update of a Outage or of a collection of Outage
- Creation of an Outage
- Deletion of an Outage

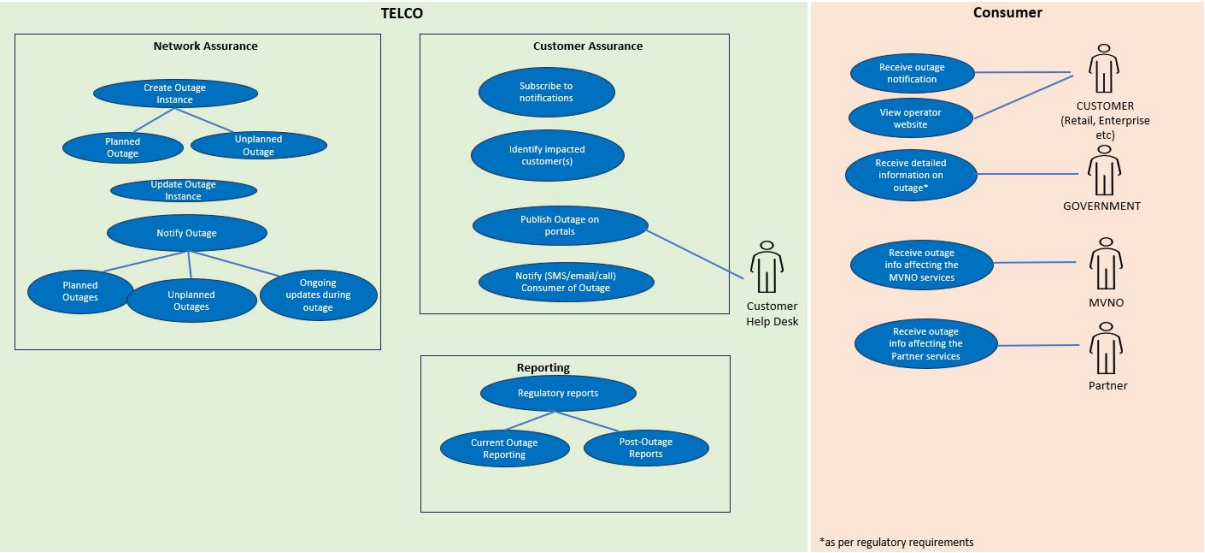
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Sample Use Cases

Once the Outage management system identifies an outage, it will generate an instance of the outage using the Outage Management API and publish API event notifications. The customer assurance component will then extract the necessary information from these notifications to inform customers via SMS, portal announcements, mobile apps, etc. This also allows MVNOs, partners, and government regulatory bodies to subscribe to notifications to manage interrupted services, respond to emergencies, and enforce regulatory compliance.

- UC1 - Declare Outage
  - Who: I'm an Outage Management component
  - When: I have received details of an unplanned or planned event, I need to declare an outage
  - What: I need to declare an outage as the unplanned or planned event is impacting customer service(s) and/or geographic areas
  - How: I will create an outage instance with the required attributes captured against the outage instance
- UC2 - Report on Outage instances
  - Who: I'm a Regulatory body, Partner, MVNO
  - When: On a need by basis
  - What: I require a list of outages during a period of time
  - How: I will query the Outage Management component to get all outage instances for a period of time
- UC3 - Subscribe to Outage notification
  - Who: I'm Customer Assurance, Partner, Regulatory, MVNO
  - When: There is an outage due to a planned or unplanned event
  - What: I want to be notified what is the impact of the outage and estimated restoration time as well as ongoing updates
  - How: I need to be able to subscribe to outage notifications from the Telco operator

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a CloudOutage and NetworkOutage inheriting properties from the abstract Outage entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Outage instances some may be instances of CloudOutage where other could be instances of NetworkOutage. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an OutageRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as CloudOutageRef or NetworkOutageRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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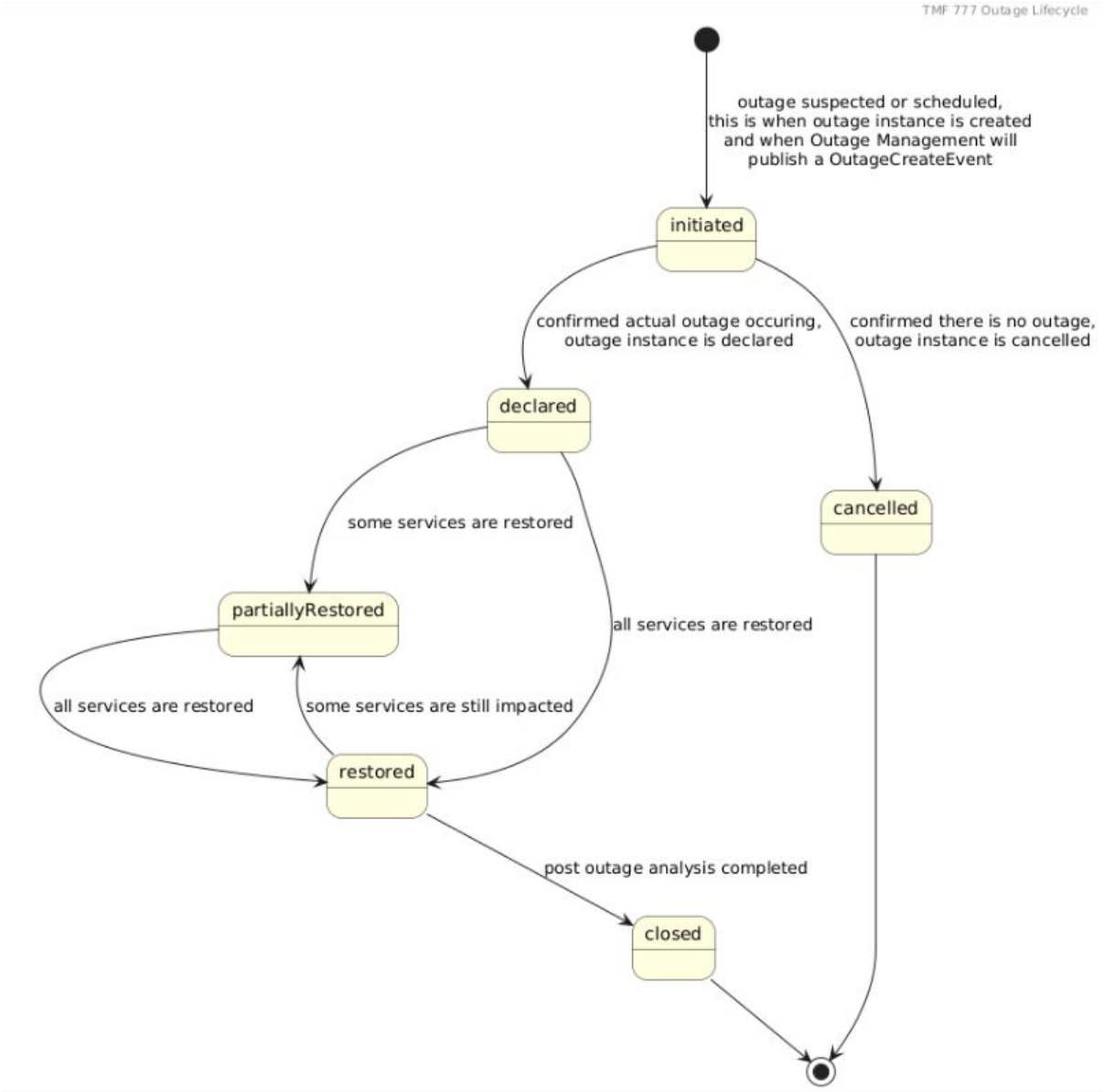
STATE LIFECYCLE

Lifecycle of Outage

The following table provided outage state explanation. The state graphic gives an overview of the allowed state changes.

State	Definition
initiated	This is an initial state of an outage and represents a suspected outage or planned outage scheduled.
cancelled	Outage cancelled as outage has been ruled out (e.g., it was a false alarm or no longer planned)
declared	Outage is occuring and must be declared.
partiallyRestored	A significant amount (e.g. at least half) of services/locations that were impacted have been restored
restored	All the services that were impacted have been restored
closed	Post outage analysis completed, and steps identified to prevent similar outages in the future

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RESOURCE MODEL

Managed Entity and Task Resource Models

Outage resource

An Outage is a formally declared period of disruption or interruption of service(s) in a specific geographical region(s), preventing users from effectively accessing or utilizing the service. This could be planned or unplanned.

Resource model

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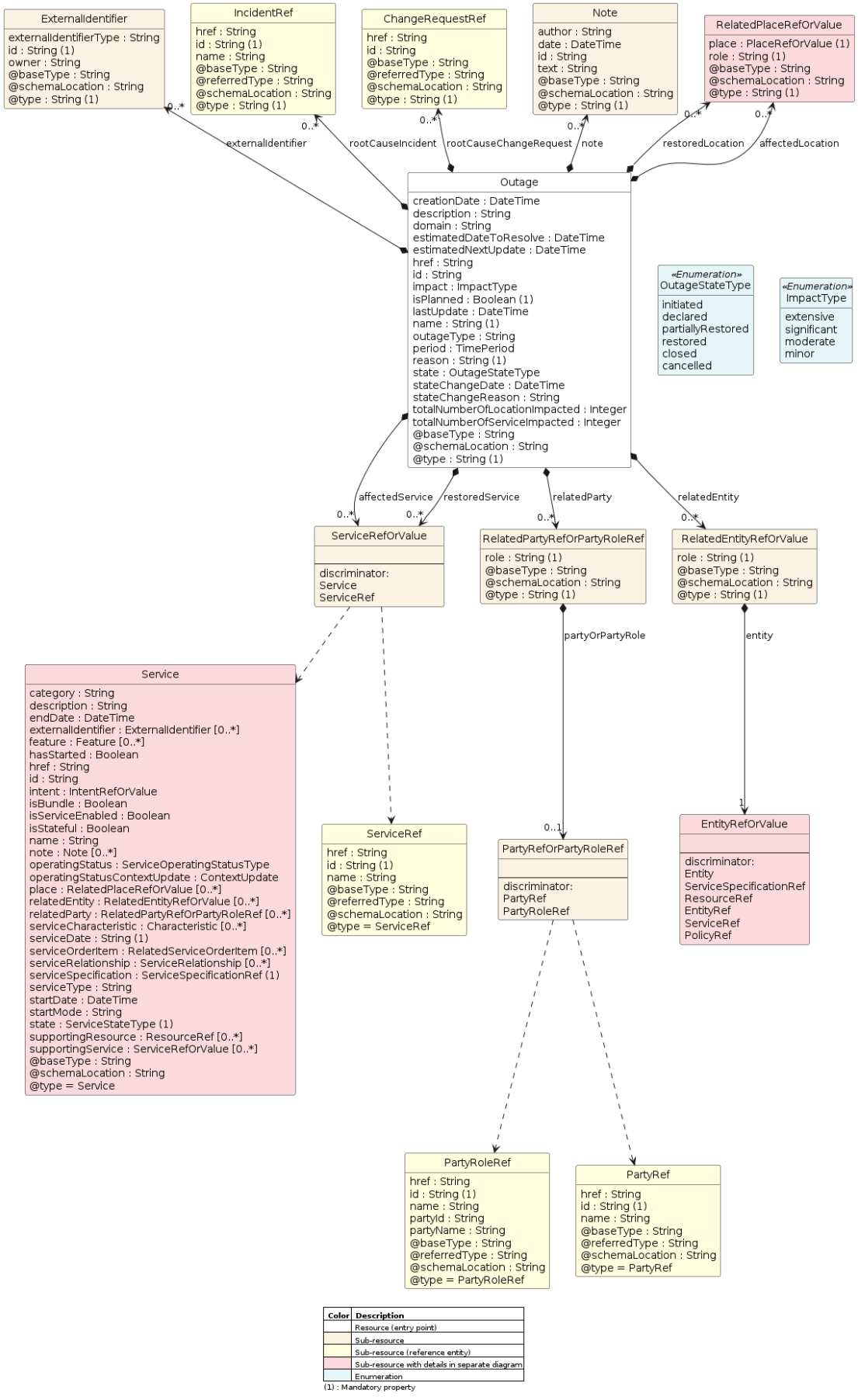
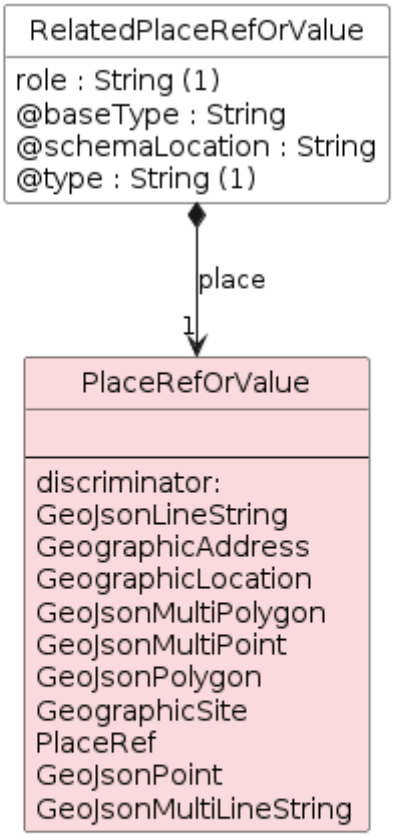


Figure 1 - Outage

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Color	Description
	Sub-resource (sub-graph in current diagram)
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 2 - RelatedPlaceRefOrValue

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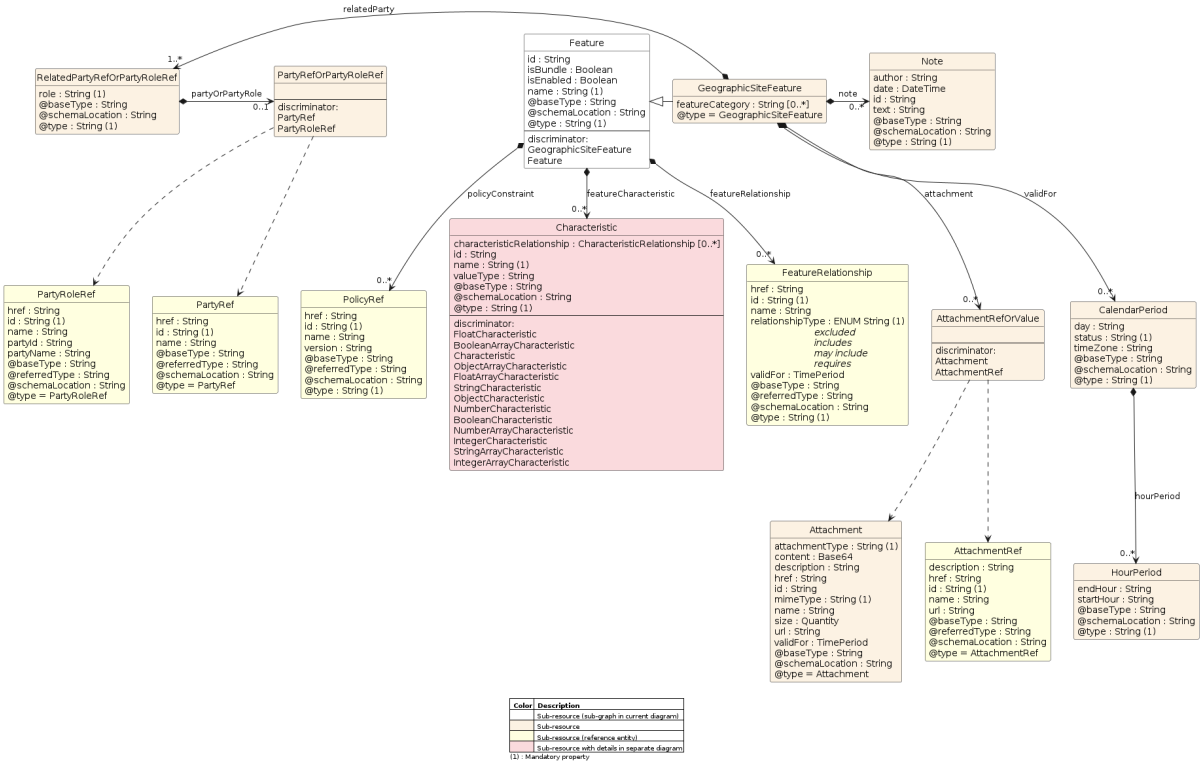


Figure 3 - Feature

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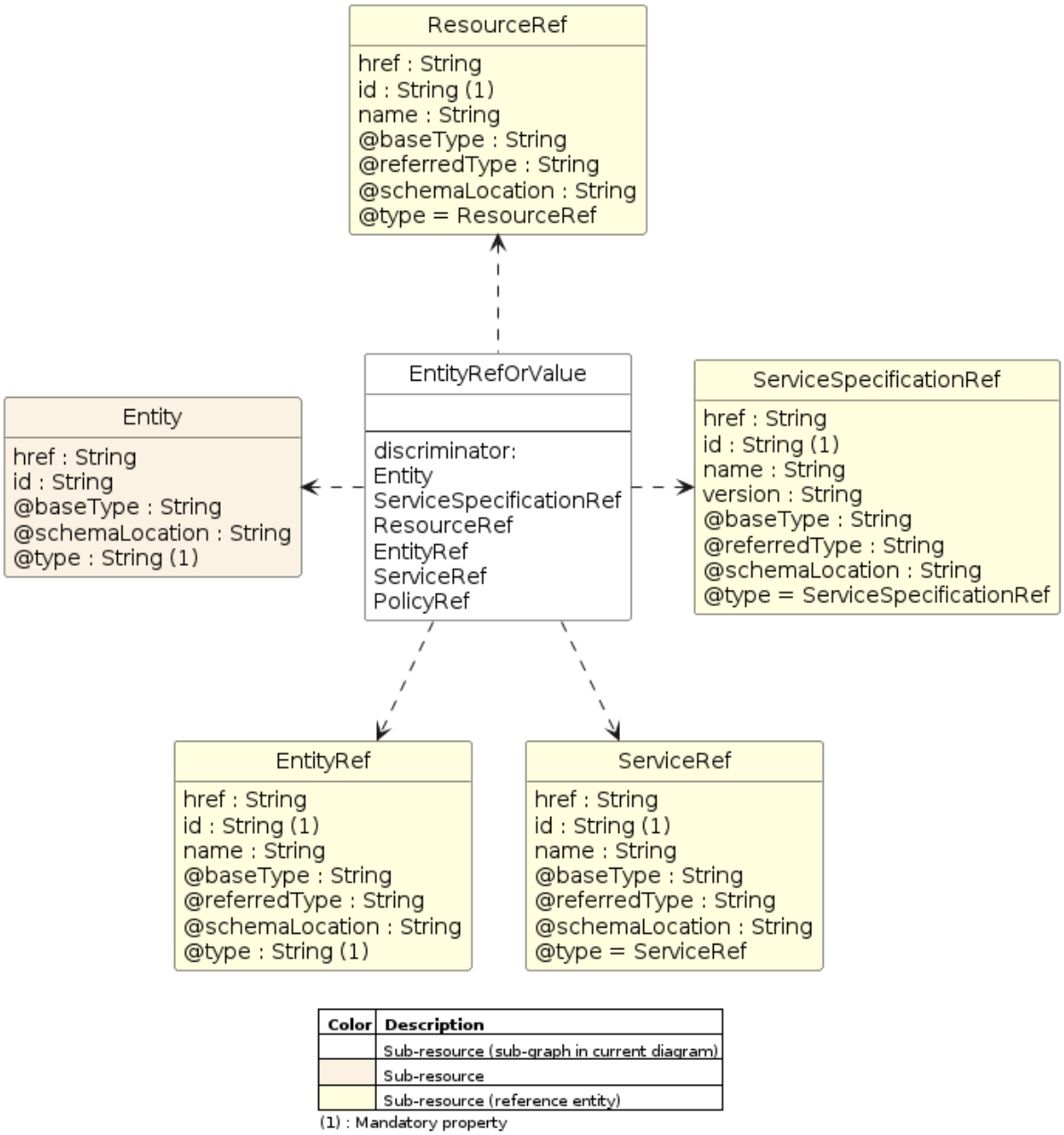


Figure 4 - EntityRefOrValue

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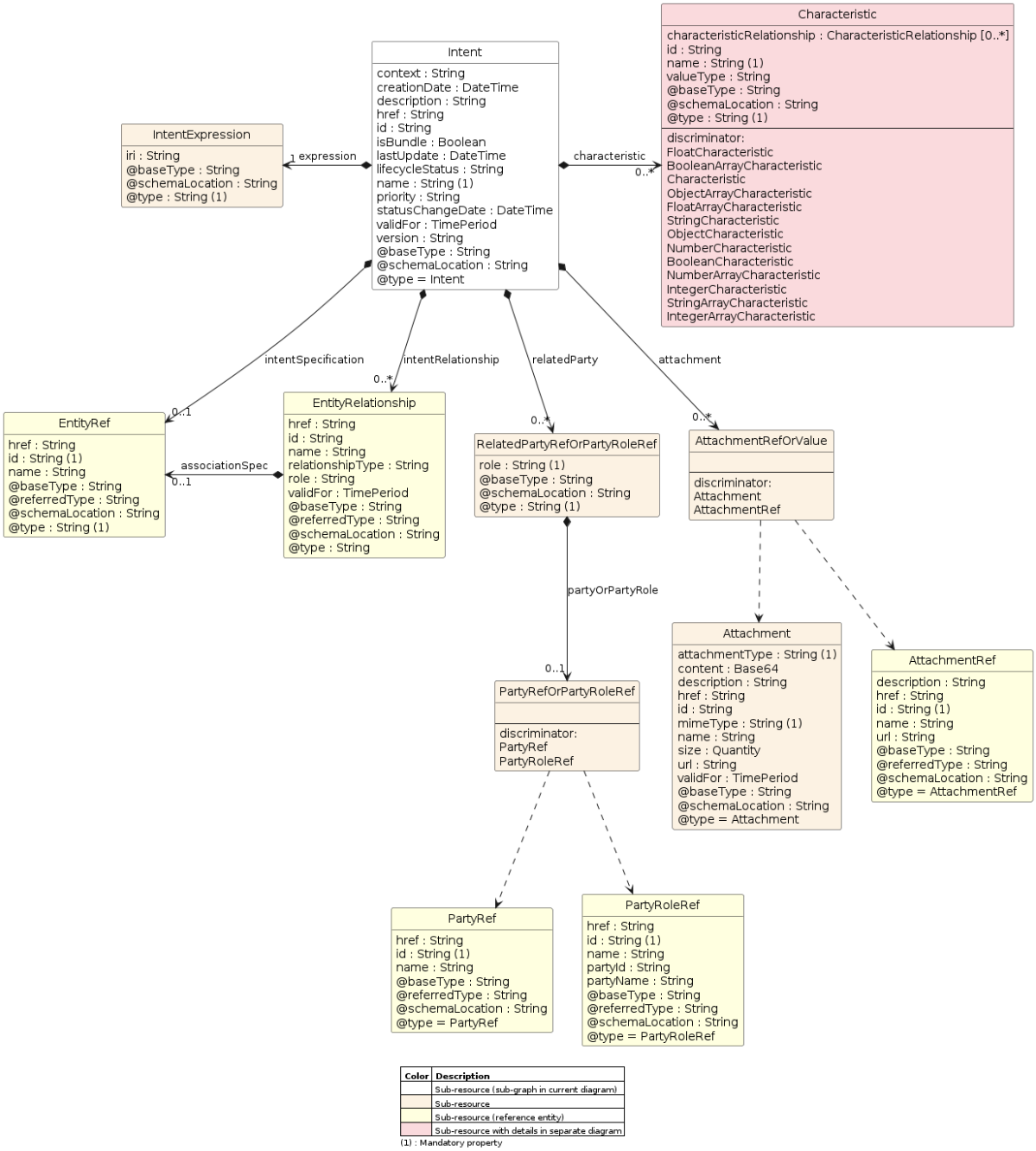


Figure 5 - Intent

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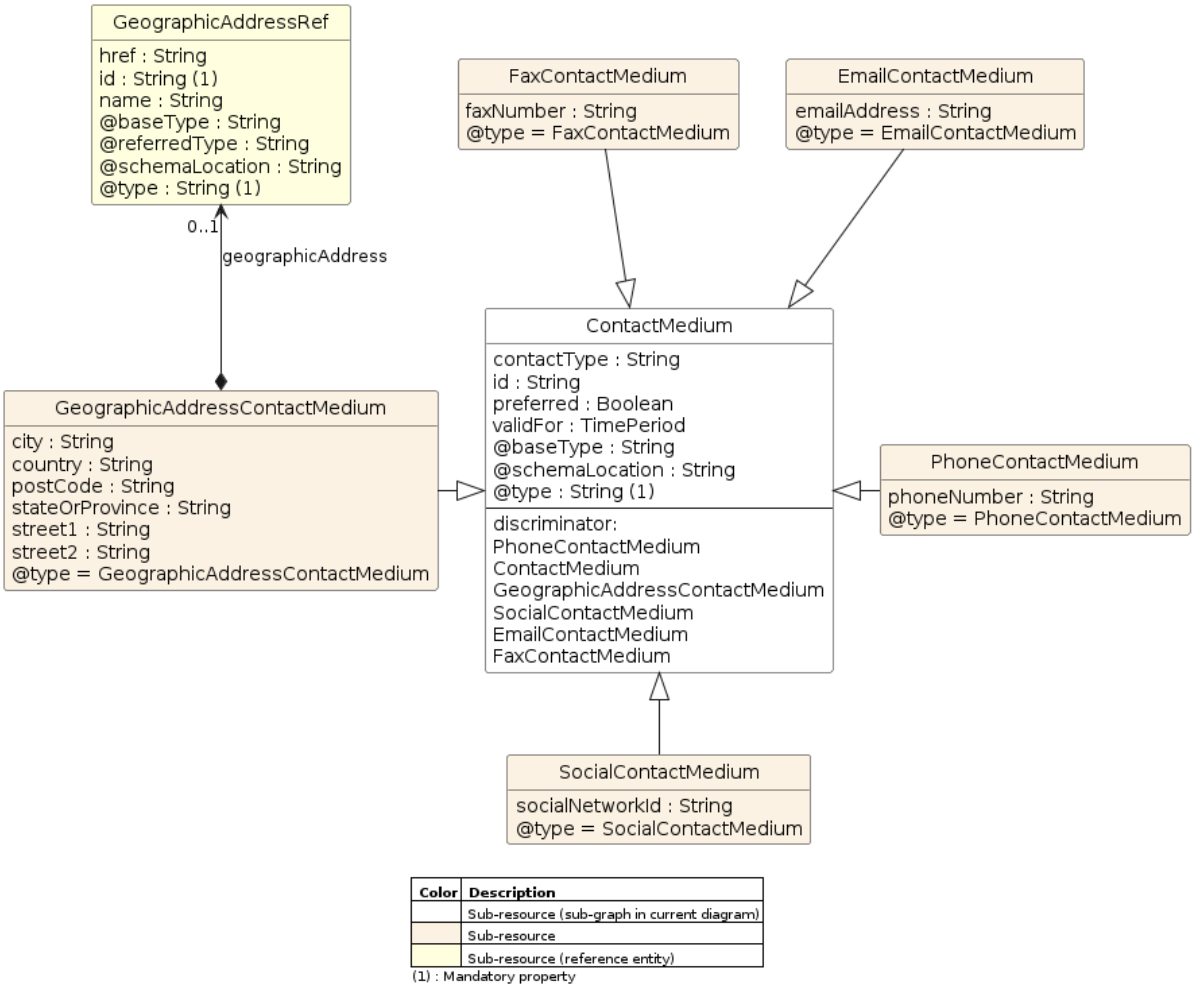


Figure 6 - ContactMedium

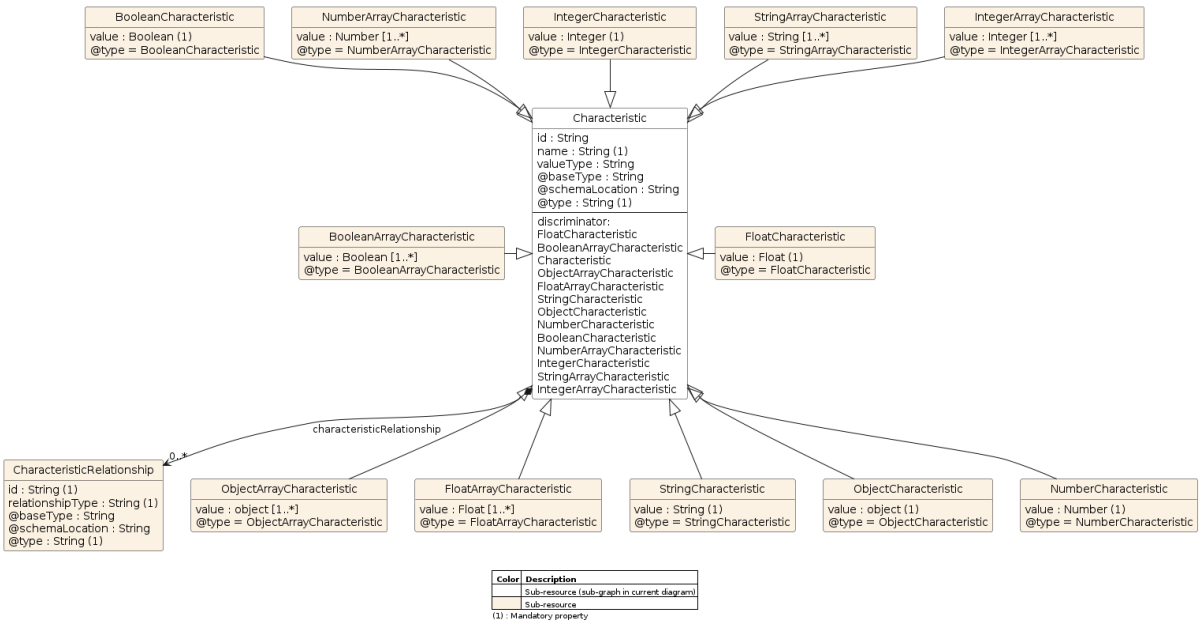


Figure 7 - Characteristic

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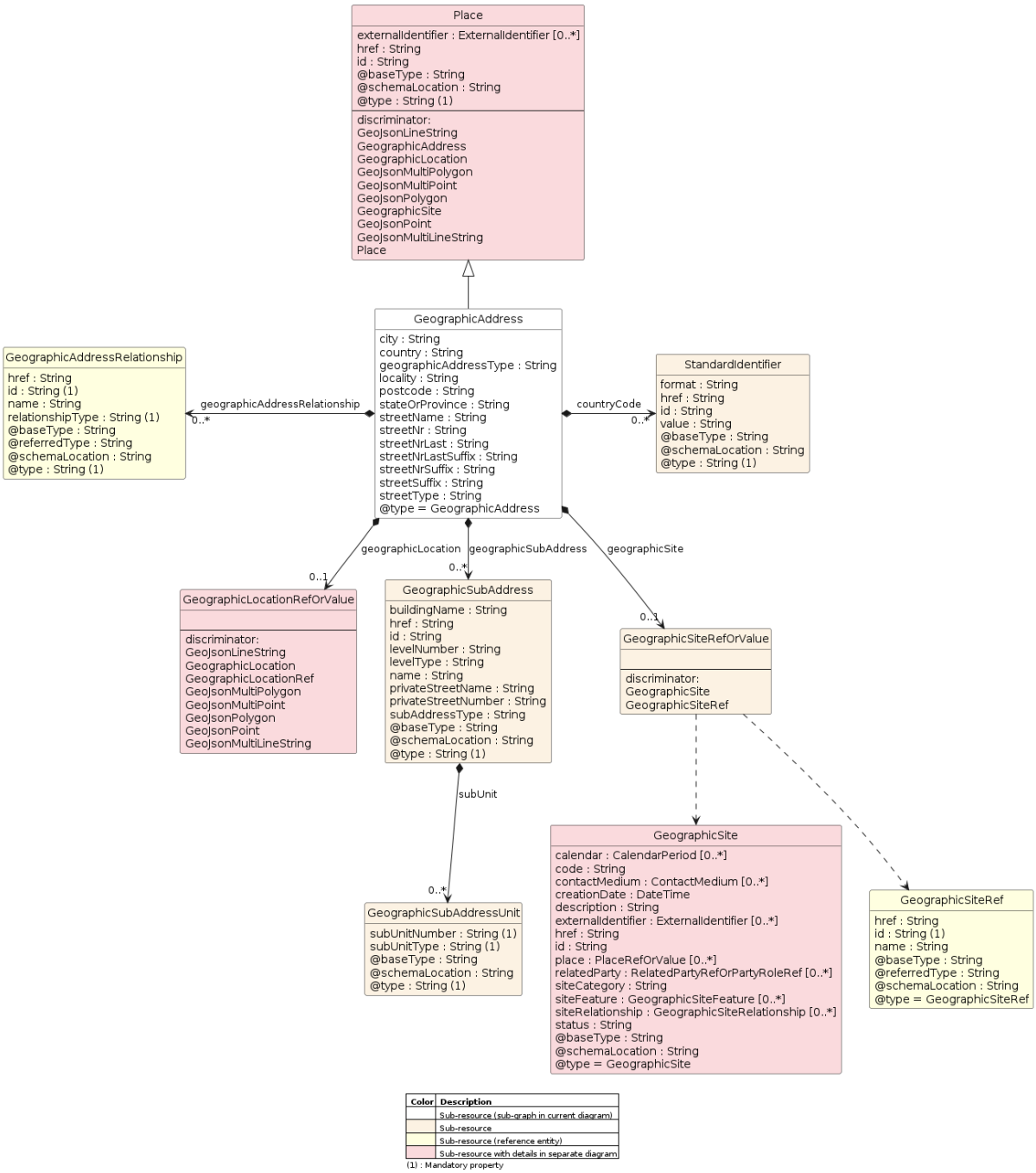


Figure 8 - GeographicAddress

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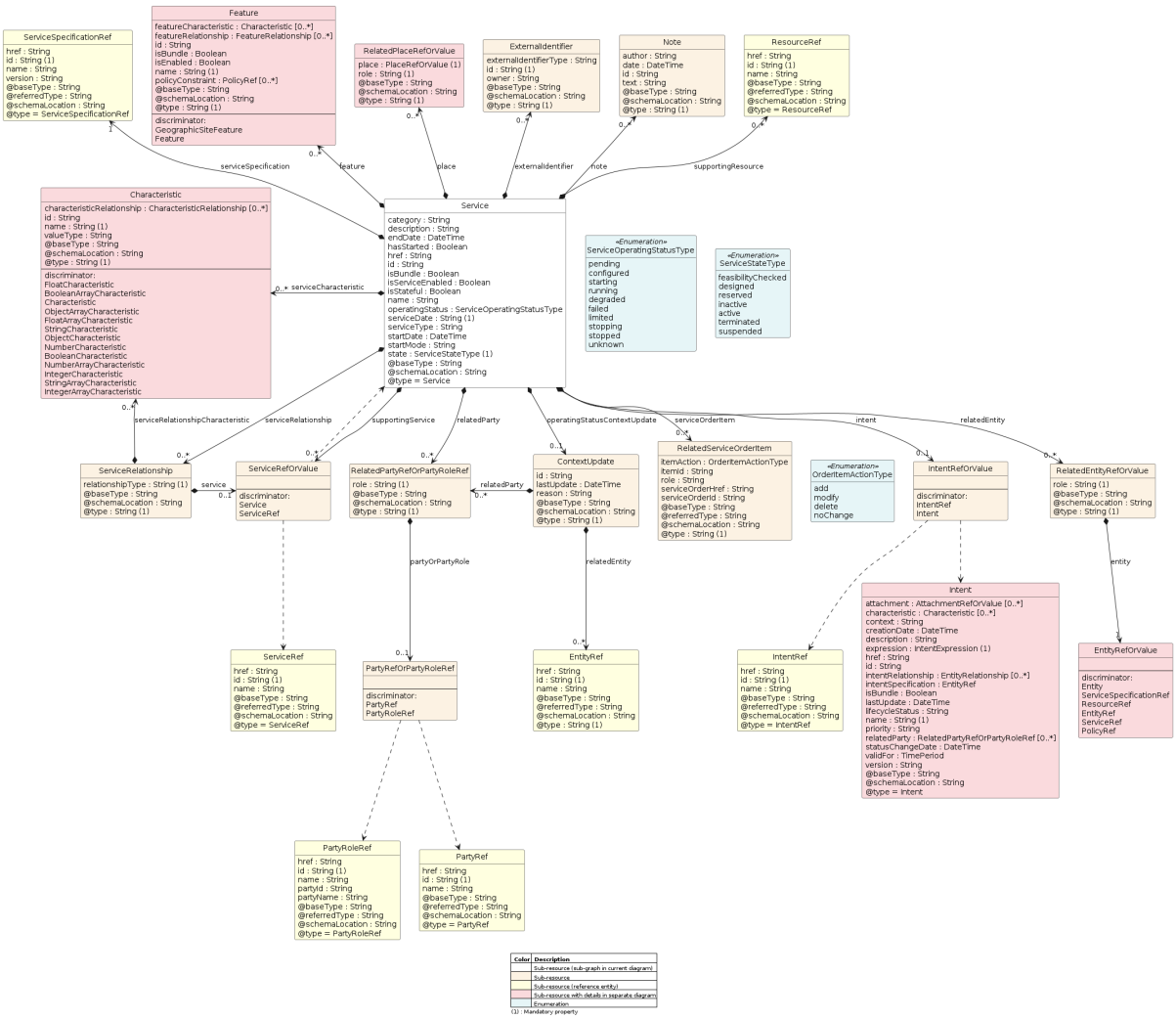


Figure 9 - Service

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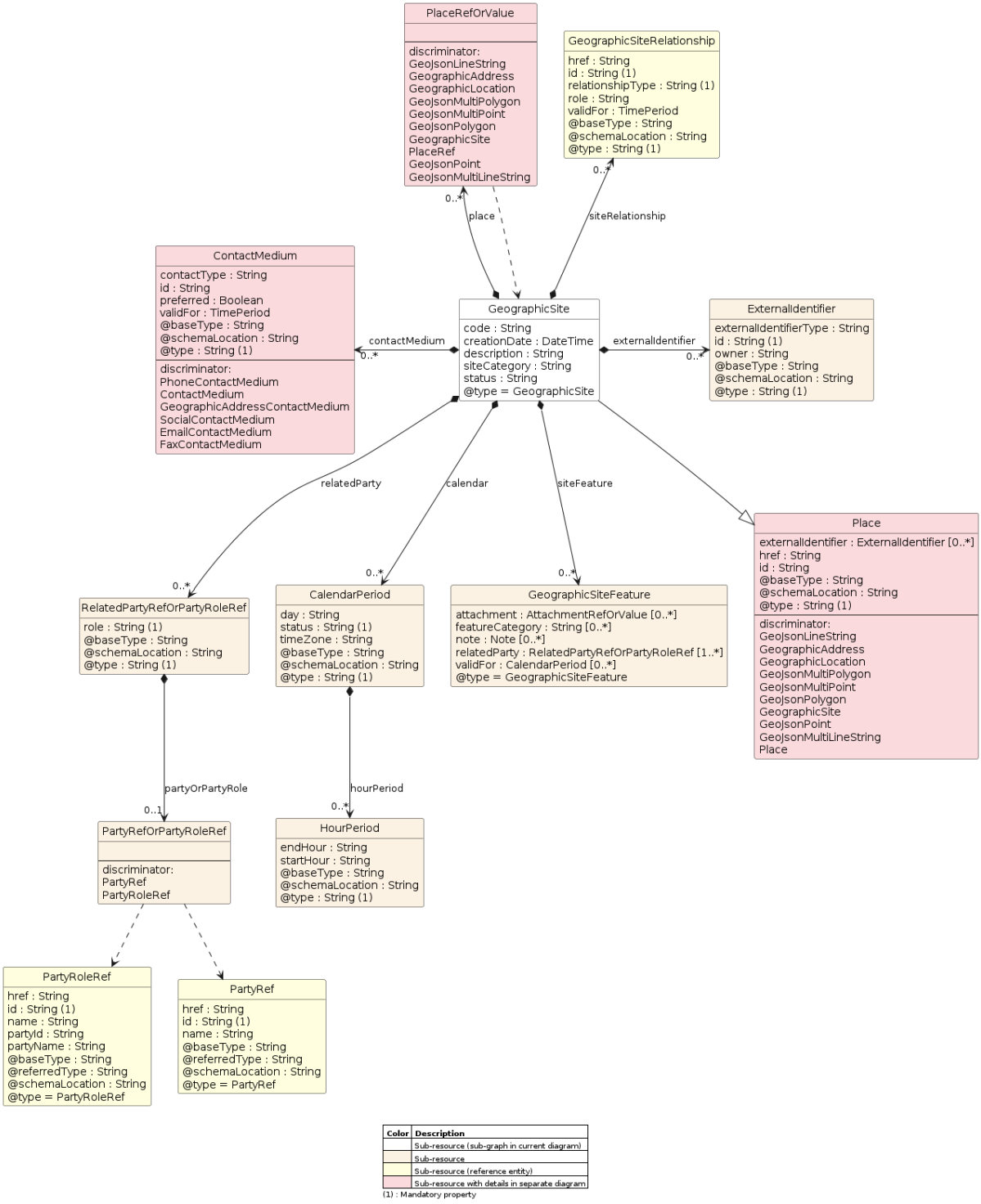


Figure 10 - GeographicSite

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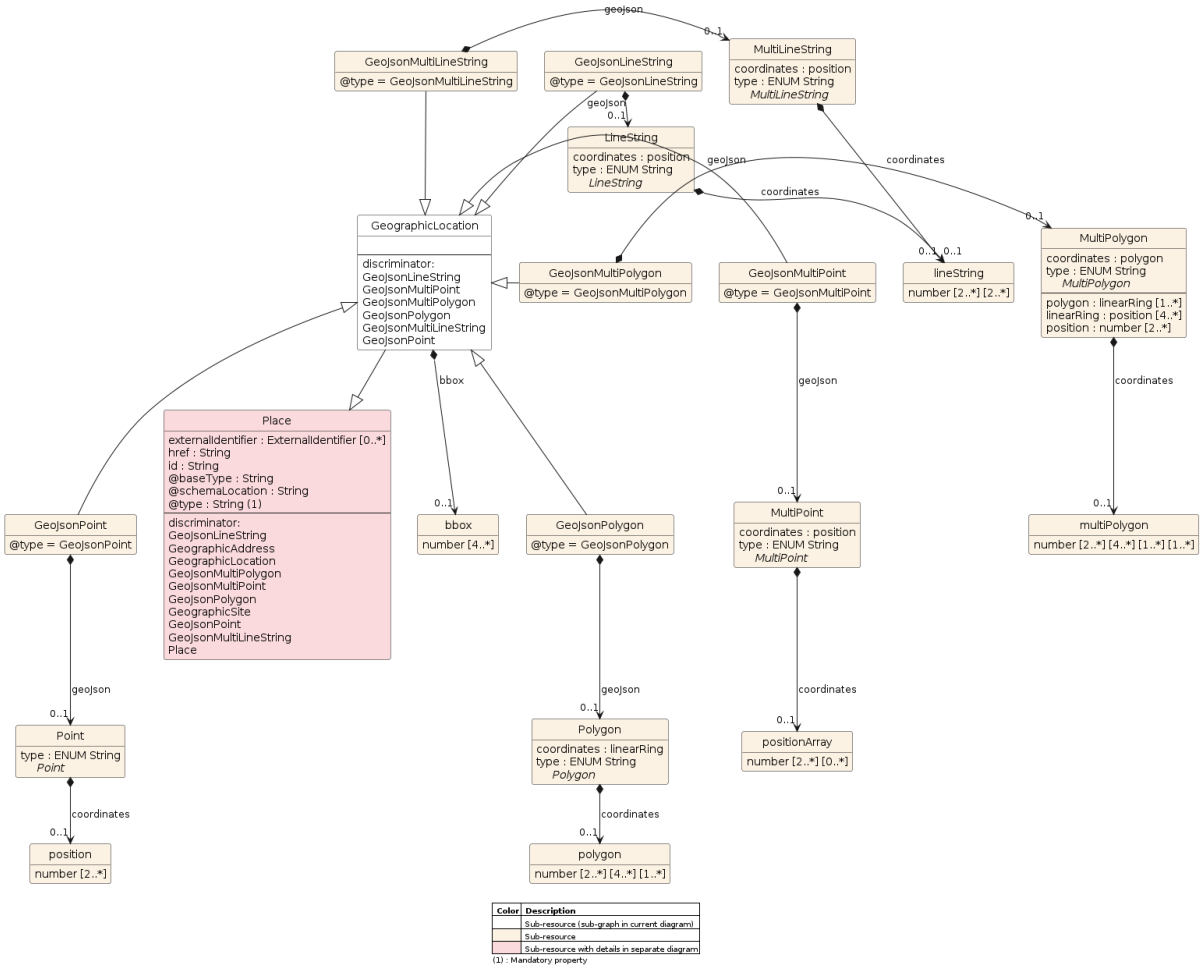


Figure 11 - GeographicLocation

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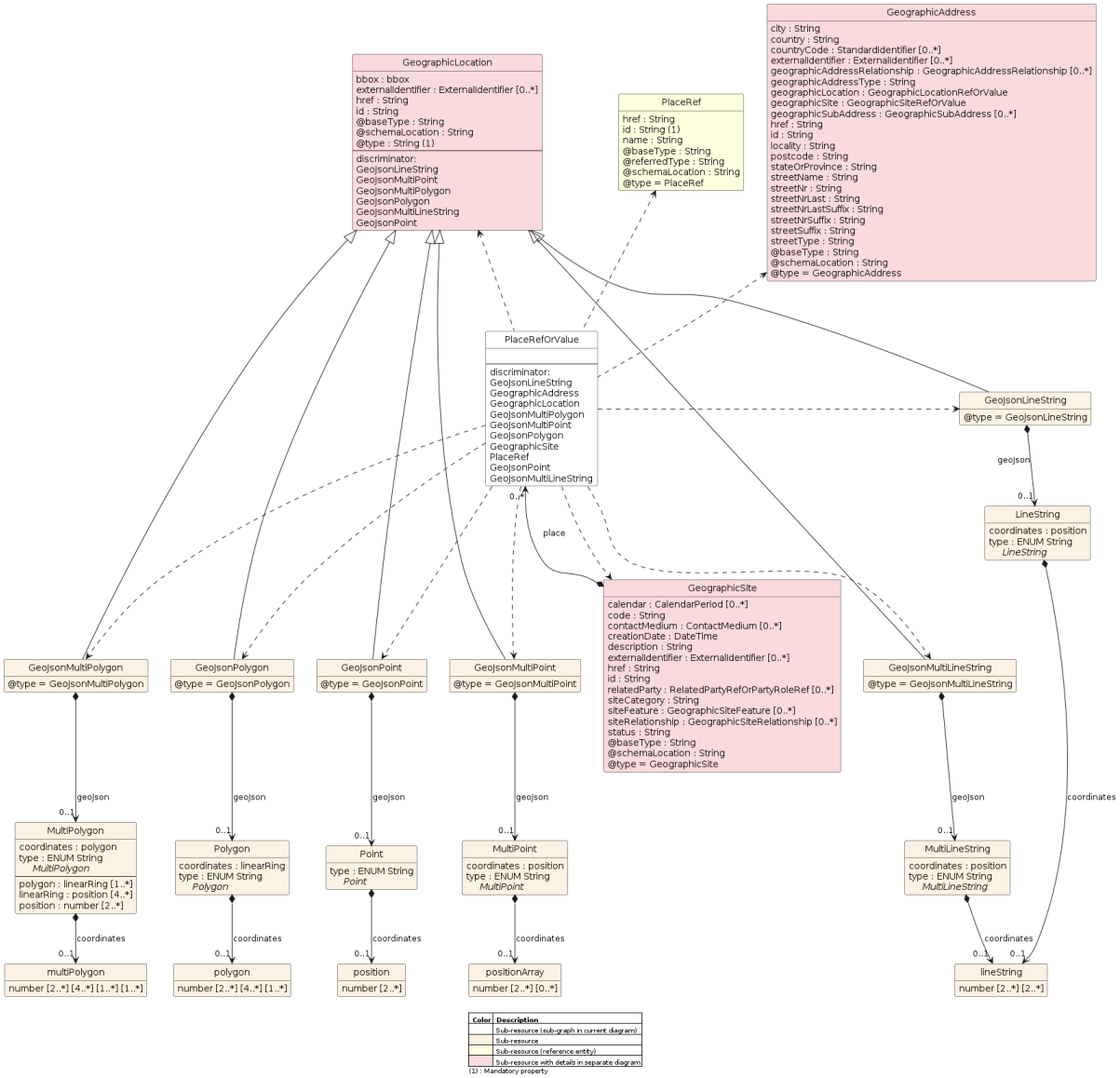


Figure 12 - PlaceRefOrValue

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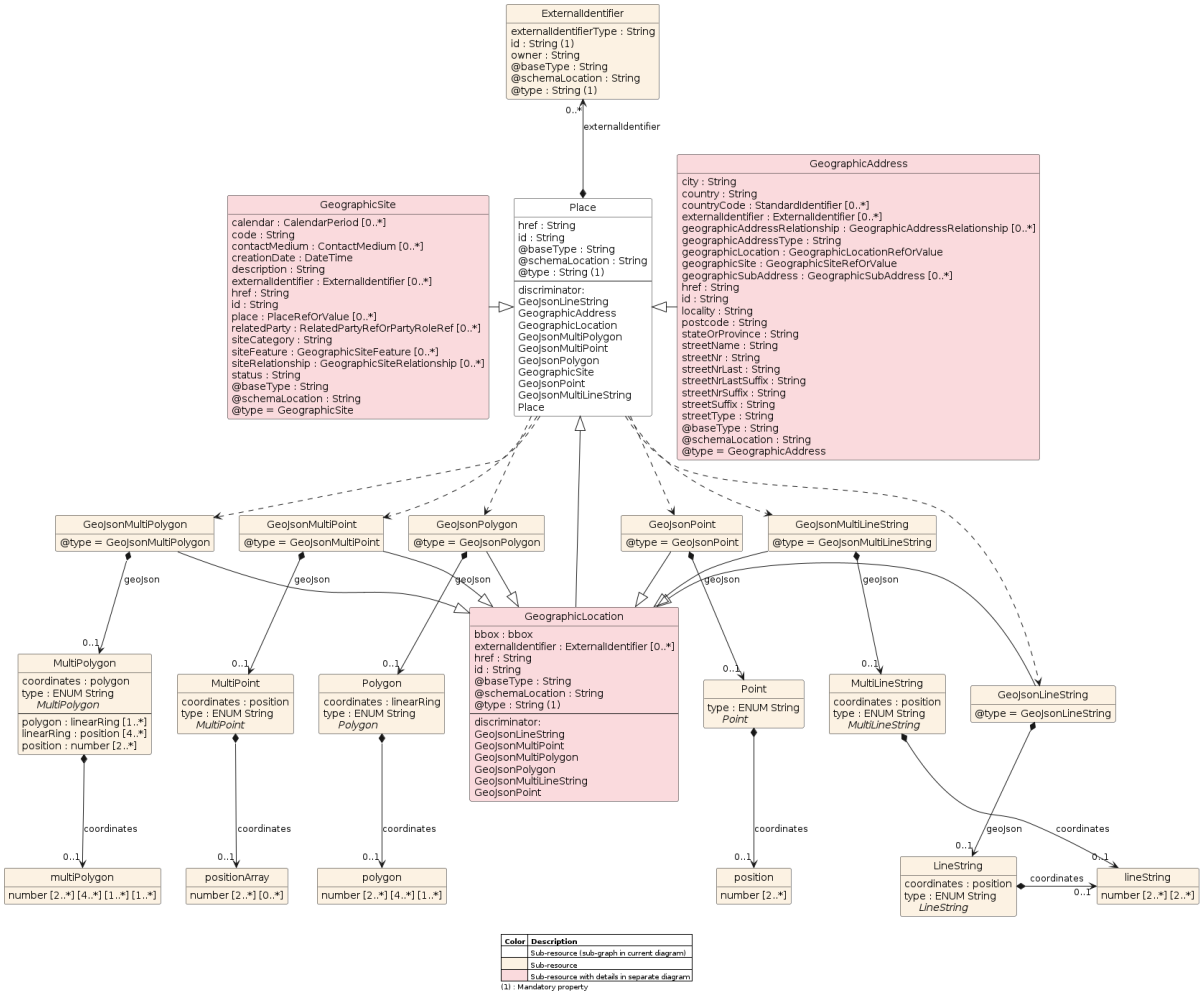


Figure 13 - Place

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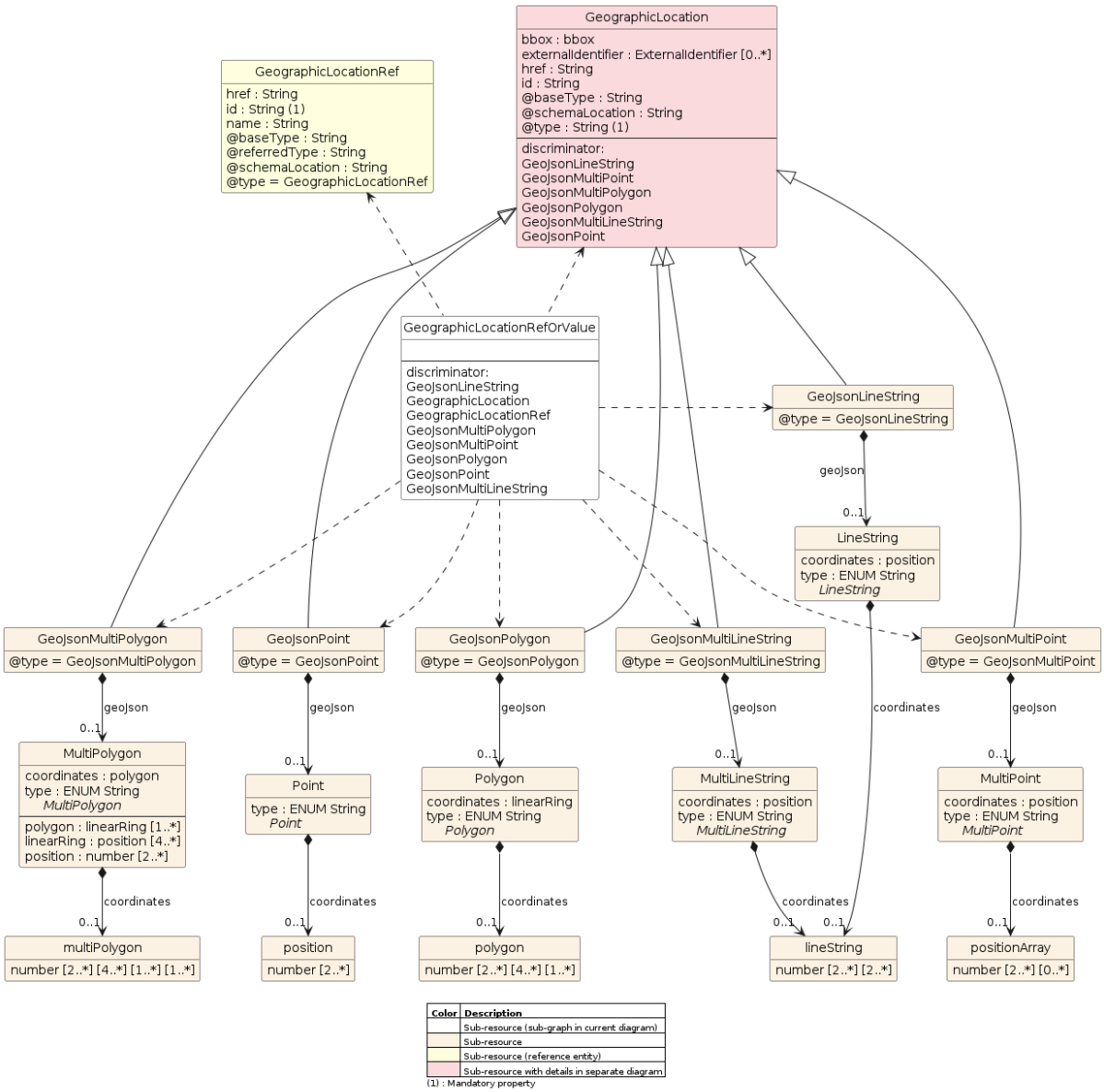


Figure 14 - GeographicLocationRefOrValue

Field descriptions

Outage fields

affectedLocation	A RelatedPlaceRefOrValue. A list of the locations affected by the outage. At least one of affectedService or affectedLocation should be present.
affectedService	A ServiceRefOrValue. List of the services affected by the outage. At least one of affectedService or affectedLocation should be present.
creationDate	A DateTime. Indicates the time (as a date + time) at which the outage was created. It indicates the time when an outage instance is created after identifiying incident or change request and recording suspected impact.

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description	A String. A textual succinct description of the nature, symptoms, cause, or effect of the outage.
domain	A String. The network domain or technology context of the outage, for example RAN, PON, OTN, Cloud, Cross-Domain etc.
estimatedDateToResolve	A DateTime. Indicates the estimated date (as a date + time) at which the outage will be resolved. That is, when all the affected service and/or locations are restored.
estimatedNextUpdate	A DateTime. The estimated time when next update will be provided.
externalIdentifier	An ExternalIdentifier. External identifier of the outage instance that is different to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
impact	An ImpactType. Possible values for the impact of an Incident. ENUMERATED with values: * extensive * significant * moderate * minor
isPlanned	A Boolean. True if outage is planned. false if outage unplanned.
lastUpdate	A DateTime. Date and time of the last update of the Outage.
name	A String. The name of the outage. A concise string or identifier that briefly distinguishes or categorizes the outage.
note	A Note. A list of comments or notes made on the outage.
outageType	A String. Describes the type of outage.
period	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
reason	A String. A textual succinct description of the cause/reason for the outage.
relatedEntity	A RelatedEntityRefOrValue. List of related entities associated with this outage. This could be cost, reporting management template etc.
relatedParty	A RelatedPartyRefOrPartyRoleRef. List of parties or party roles playing a role within the outage.
restoredLocation	A RelatedPlaceRefOrValue. A list of the locations restored and no longer affected by the outage.
restoredService	A ServiceRefOrValue. List of the services restored and no longer affected by the outage.
rootCauseChangeRequest	A ChangeRequestRef. List of change request(s) that are associated to the outage. At least one of rootCauseChangeRequest or rootCauseIncident should be present.

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rootCauseIncident	An IncidentRef. List of incident(s) that are associated to the outage. At least one of rootCauseChangeRequest or rootCauseIncident should be present.
state	An OutageStateType. Possible values for the state of an Outage. ENUMERATED with values: * initiated * declared * partiallyRestored * restored * closed * cancelled
stateChangeDate	A DateTime. Time the state was last changed.
stateChangeReason	A String. The reason of state change.
totalNumberOfLocationImpacted	An Integer. Once the location impact is finalised, this will represent the total number of locations impacted.
totalNumberOfServiceImpacted	An Integer. Once the service impact is finalised, this will represent the total number of services impacted.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeoJsonLineString sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A LineString. GeoJSON: A collection of Points forming a connected line.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LineString sub-resource fields

coordinates	A position. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * LineString

GeoJsonMultiLineString sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiLineString. GeoJSON: A collection of distinct LineStrings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiLineString sub-resource fields

coordinates	A position. GeoJSON: An array of two or more positions.
type	A String. ENUMERATED with values: * MultiLineString

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GeoJsonMultiPoint sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiPoint. GeoJSON: A collection of Points.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiPoint sub-resource fields

coordinates	A position. GeoJSON: An array of positions.
type	A String. ENUMERATED with values: * MultiPoint

GeoJsonMultiPolygon sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A MultiPolygon. GeoJSON: A collection of Polygons.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MultiPolygon sub-resource fields

coordinates	A polygon. GeoJSON: An array of polygons.
type	A String. ENUMERATED with values: * MultiPolygon

GeoJsonPoint sub-resource fields

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bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A Point. GeoJSON: A single position.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Point sub-resource fields

coordinates	A position. GeoJSON: A single position.
type	A String. ENUMERATED with values: * Point

GeoJsonPolygon sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geoJson	A Polygon. GeoJSON: An array of linear rings.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Polygon sub-resource fields

coordinates	A linearRing. GeoJSON: An array of linear rings.
type	A String. ENUMERATED with values: * Polygon

GeographicAddress sub-resource fields

city	A String. City that the address is in.
country	A String. Country that the address is in.

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countryCode	A StandardIdentifier. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
geographicAddressRelationship	A GeographicAddressRelationship. A list of geographic addresses relationships. Represents a relationship between geographic addresses.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress. Collection of sub addresses.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	GeographicLocation can be instantiated as * GeoJsonLineString * GeoJsonMultiLineString * GeoJsonMultiPoint * GeoJsonMultiPolygon * GeoJsonPoint * GeoJsonPolygon
geoJson	This property is present in subclasses

GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod. Collection of calendar items.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
contactMedium	A ContactMedium. Collection of contact information. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.
externalIdentifier	An ExternalIdentifier. Collection of external identifiers.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
place	A PlaceRefOrValue. Collection of place objects.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Related parties collection.
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature. Collection of site features.
siteRelationship	A GeographicSiteRelationship. Collection of site siteRelationships.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, …).
hourPeriod	A HourPeriod. Collection of hour intervals.
status	A String. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

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contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium
city	This property is present in subclasses
country	This property is present in subclasses
emailAddress	This property is present in subclasses
faxNumber	This property is present in subclasses
geographicAddress	This property is present in subclasses
phoneNumber	This property is present in subclasses
postCode	This property is present in subclasses
socialNetworkId	This property is present in subclasses
stateOrProvince	This property is present in subclasses
street1	This property is present in subclasses
street2	This property is present in subclasses

GeographicAddressRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

EmailContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressContactMedium sub-resource fields

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.

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id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

SocialContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteFeature sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this specification, such as picture, document, media.
featureCategory	A String. Collection of feature category.
featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
note	A Note. Collection of notes.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
relatedParty	A RelatedPartyRefOrPartyRoleRef. The associated party or partyRole.
validFor	A CalendarPeriod. Collection of validity information.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

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attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. A characteristic which value is an array of Float(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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relationshipType	A String. This is the type of the feature relationship. ENUMERATED with values: * excluded * includes * may include * requires
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

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featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
featureRelationship	A FeatureRelationship. Collection of feature relationships.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to manage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Feature can be instantiated as * GeographicSiteFeature
attachment	This property is present in subclasses
featureCategory	This property is present in subclasses
note	This property is present in subclasses
relatedParty	This property is present in subclasses
validFor	This property is present in subclasses

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.
id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

See [GeographicLocation] for details.

GeographicSite sub-resource fields

See [GeographicSite] for details.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Service sub-resource fields

category	A String. Is it a customer facing or resource facing service.
description	A String. Free-text description of the service.
endDate	A DateTime. Date when the service ends.
externalIdentifier	An ExternalIdentifier. A list of external identifiers assoicated with this service.

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feature	A Feature. A list of feature associated with this service. Feature can be instantiated as * GeographicSiteFeature
hasStarted	A Boolean. If TRUE, this Service has already been started.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRefOrValue. Intent Ref (if Intent already exists) or Value (if Intent be created or its details be presented).
isBundle	A Boolean. If true, the service is a ServiceBundle which regroup a service hierachy. If false, the service is a 'atomic' service (hierachy leaf).
isServiceEnabled	A Boolean. If FALSE and hasStarted is FALSE, this particular Service has NOT been enabled for use - if FALSE and hasStarted is TRUE then the service has failed.
isStateful	A Boolean. If TRUE, this Service can be changed without affecting any other services.
name	A String. Name of the service.
note	A Note. A list of notes made on this service.
operatingStatus	A ServiceOperatingStatusType. Valid values for the Operating status of the service. ENUMERATED with values: * pending * configured * starting * running * degraded * failed * limited * stopping * stopped * unknown
operatingStatusContextUpdate	A ContextUpdate. Holds the date when the status/state last changed and provides a reason/context to the current value of the status/state. It has the relatedEntity and relatedParty in relation to the status/state value.
place	A RelatedPlaceRefOrValue. A list of places (Place [*]). Used to define a place useful for the service (for example a geographical place whre the service is installed).
relatedEntity	A RelatedEntityRefOrValue. A list of related entities in relationship with this service.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A list of related party references (RelatedParty [*]). A related party defines party or party role linked to a specific entity.

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serviceCharacteristic	A Characteristic. A list of characteristics that characterize this service (ServiceCharacteristic [*]). Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
serviceDate	A String. Date when the service was created (whatever its status).
serviceOrderItem	A RelatedServiceOrderItem. A list of service order items related to this service.
serviceRelationship	A ServiceRelationship. A list of service relationships (ServiceRelationship [*]). Describes links with other service(s) in the inventory.
serviceSpecification	A ServiceSpecificationRef. Service specification reference: ServiceSpecification(s) required to realize a ProductSpecification.
serviceType	A String. Business type of the service.
startDate	A DateTime. Date when the service starts.
startMode	A String. This attribute is an enumerated integer that indicates how the Service is started, such as: 0: Unknown; 1: Automatically by the managed environment; 2: Automatically by the owning device; 3: Manually by the Provider of the Service; 4: Manually by a Customer of the Provider; 5: Any of the above.
state	A ServiceStateType. Valid values for the lifecycle state of the service. ENUMERATED with values: * feasibilityChecked * designed * reserved * inactive * active * terminated * suspended
supportingResource	A ResourceRef. A list of supporting resources (SupportingResource [*]).Note: only Service of type RFS can be associated with Resources.
supportingService	A ServiceRefOrValue. A list of supporting services (SupportingService [*]). A collection of services that support this service (bundling, link CFS to RFS).
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Intent sub-resource fields

attachment	An AttachmentRefOrValue. Attachments that may be of relevance to this intent, such as picture, document, media.
characteristic	A Characteristic. Collection of characteristics. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
context	A String. A string used to give a context to the intent.
creationDate	A DateTime. Date and time of the creation of this REST resource.
description	A String. The description of the intent.
expression	An IntentExpression. An Intent expression.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentRelationship	An EntityRelationship. A list of intents related to this intent.
intentSpecification	An EntityRef. Entity reference schema to be use for all entityRef class.
isBundle	A Boolean. IsBundle determines whether an intent represents a single intent (false), or a bundle of intents(true).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this intent.
name	A String. The name of the intent.
priority	A String. Can be used by intent owner to prioritize intents in an intent management system.
relatedParty	A RelatedPartyRefOrPartyRoleRef. Collection of related parties.

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statusChangeDate	A DateTime. A date time(DateTime). The date that the entity status changed to the current one.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. A field that identifies the specific version of an instance of an intent.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentExpression sub-resource fields

iri	A String. Internationalized Resource Identifier of the intent Expression.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRelationship sub-resource fields

associationSpec	An EntityRef. Entity reference schema to be use for all entityRef class.
href	A String.
id	A String.
name	A String.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String.

EntityRef sub-resource fields

href	A String. Hyperlink reference.

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id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

See [EntityRef] for details.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContextUpdate sub-resource fields

id	A String. Identifier of the context update.
lastUpdate	A DateTime. The date and time the status/state last changed.
reason	A String. The reason/context for the current value of the status/state.
relatedEntity	An EntityRef. A list of entities that represent observations, anomalies or managing entities etc associated to the current status/state value.
relatedParty	A RelatedPartyRefOrPartyRoleRef. List of parties associated to the current status/state value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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RelatedEntityRefOrValue sub-resource fields

entity	An EntityRefOrValue.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Service specification version.

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@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedServiceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the product. ENUMERATED with values: * add * modify * delete * noChange
itemId	A String. Identifier of the order item where the service was managed.
role	A String. Role of the service order item for this service.
serviceOrderHref	A String. Reference of the related entity.
serviceOrderId	A String. Unique identifier of a related entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceRelationship sub-resource fields

relationshipType	A String.
service	A ServiceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Service entity and not the ServiceRefOrValue class itself.

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serviceRelationshipCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatArrayCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ServiceSpecificationRef sub-resource fields

See [ServiceSpecificationRef] for details.

ResourceRef sub-resource fields

See [ResourceRef] for details.

ServiceRef sub-resource fields

See [ServiceRef] for details.

ExternalIdentifier sub-resource fields

See [ExternalIdentifier] for details.

Note sub-resource fields

See [Note] for details.

RelatedEntityRefOrValue sub-resource fields

See [RelatedEntityRefOrValue] for details.

RelatedPartyRefOrPartyRoleRef sub-resource fields

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See [RelatedPartyRefOrPartyRoleRef] for details.

RelatedPlaceRefOrValue sub-resource fields

See [RelatedPlaceRefOrValue] for details.

ChangeRequestRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IncidentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'Outage' resource object.


```
{
  "@type": "Outage",
  "@schemaLocation": "http://my.schemas/Outage.schema.json",
  "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-456",
  "id": "123-456",
  "name": "QWR23 FDH down",
  "description": "broadband outage due to software failure at FDH",
  "isPlanned": false,
  "domain": "broadband",
  "reason": "software failure at FDH",
  "state": "initiated",
  "stateChangeDate": "2025-04-23T04:16:56Z",
  "stateChangeReason": "identified from INC-789789",
```


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```
  "impact": "significant",
  "outageType": "significant local",
  "creationDate": "2025-04-23T04:16:56Z",
  "estimatedDateToResolve": "2025-04-23T10:16:56Z",
  "estimatedNextUpdate": "2025-04-23T05:16:56Z",
  "period": {
    "startDateTime": "2025-04-23T04:16:56Z"
  },
  "rootCauseIncident": [
    {
      "@type": "IncidentRef",
      "@schemaLocation": "http://my.schemas/Incident.schema.json",
      "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
      "id": "789789",
      "name": "Incident - 59840 software failure",
      "@referredType": "Incident"
    }
  ],
  "affectedLocation": [
    {
      "@type": "RelatedPlaceRefOrValue",
      "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
      "role": "localDistributionPoint",
      "place": {
        "@type": "PlaceRef",
        "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
        "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
        "id": "2435",
        "name": "FDH location",
        "@referredType":"GeographicLocation"
      }
    }
  ],
  "totalNumberOfServiceImpacted": 3,
  "totalNumberOfLocationImpacted": 1,
  "affectedService": [
    {
      "@type": "ServiceRef",
      "@schemaLocation": "http://my.schemas/Service.schema.json",
      "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
      "id": "123699",
      "name": "broadband connectivity service",
      "@referredType":"Service"
    },
    {
      "@type": "ServiceRef",
      "@schemaLocation": "http://my.schemas/Service.schema.json",
      "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
      "id": "123700",
      "name": "broadband connectivity service",
      "@referredType":"Service"
    },
    {
      "@type": "ServiceRef",
      "@schemaLocation": "http://my.schemas/Service.schema.json",
      "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
      "id": "123701",
      "name": "broadband connectivity service",
      "@referredType":"Service"
```


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```
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
      "role": "restorationCoordinator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
        "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
        "id": "9866",
        "name": "NOC - 3494839",
        "@referredType": "Organisation"
      }
    }
  ],
  "relatedEntity": [
    {
      "@type": "RelatedEntityRefOrValue",
      "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
      "role": "associated cost",
      "entity": {
        "@type": "EntityRef",
        "@schemaLocation": "http://my.schemas/Entity.schema.json",
        "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
        "id": "9866",
        "@referredType":"ProjectedCost"
      }
    }
  ],
  "note": [
    {
      "@type": "Note",
      "@schemaLocation": "http://my.schemas/Note.schema.json",
      "id": "789789789",
      "author": "NOC operator",
      "date": "2025-04-23T04:16:56Z",
      "text": "Onsite restoration may be required"
    }
  ],
  "lastUpdate": "2025-04-23T04:16:56Z",
  "externalIdentifier": [
    {
      "@type": "ExternalIdentifier",
      "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
      "owner": "regulatory",
      "externalIdentifierType": "regReport",
      "id": "REGREPORT-124564879"
    }
  ]
}
```


Notification Resource Models

4 notifications are defined for this API.

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Notifications related to Outage:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

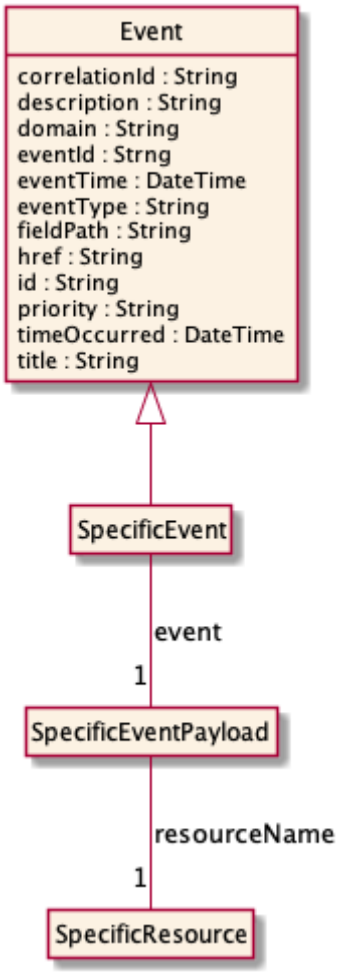


Figure 15 Notification Pattern

Outage

Create Event

Message example for OutageCreateEvent event

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```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "OutageCreateEvent illustration",
    "domain": "Network Assurance",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2025-04-23T04:16:56.176Z",
    "eventType": "OutageCreateEvent",
    "priority": "5",
    "timeOccurred": "2025-04-23T04:16:56.176Z",
    "title": "OutageCreateEvent",
    "event": {
      "outage": {
        "@type": "Outage",
        "@schemaLocation": "http://my.schemas/Outage.schema.json",
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
456",
        "id": "123-456",
        "name": "QWR23 FDH down",
        "description": "broadband outage due to software failure at FDH",
        "isPlanned": false,
        "domain": "broadband",
        "reason": "software failure at FDH",
        "state": "initiated",
        "stateChangeDate": "2025-04-23T04:16:56Z",
        "stateChangeReason": "identified from INC-789789",
        "impact": "significant",
        "outageType": "significant local",
        "creationDate": "2025-04-23T04:16:56Z",
        "estimatedNextUpdate": "2025-04-23T05:16:56Z",
        "period": {
          "startDateTime": "2025-04-23T04:16:56Z"
        },
        "rootCauseIncident": [
          {
            "@type": "IncidentRef",
            "@schemaLocation": "http://my.schemas/Incident.schema.json",
            "href": "http://serverlocation:port/tmf-
api/Incident/v4/incident/789789",
            "id": "789789",
            "name": "Incident - 59840 software failure",
            "@referredType": "Incident"
          }
        ],
        "affectedLocation": [
          {
            "@type": "RelatedPlaceRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedPlaceRefOrValue.schema.json",
            "role": "localDistributionPoint",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
              "id": "2435",
              "name": "FDH location",
              "@referredType": "GeographicLocation"
            }
          }
        ],
        "totalNumberOfServiceImpacted": 3,
```

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```
        "totalNumberOfLocationImpacted": 1,
        "affectedService": [
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
            "id": "123699",
            "name": "broadband connectivity service",
            "@referredType": "Service"
          },
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
            "id": "123700",
            "name": "broadband connectivity service",
            "@referredType": "Service"
          },
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
            "id": "123701",
            "name": "broadband connectivity service",
            "@referredType": "Service"
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
            "role": "restorationCoordinator",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
              "id": "9866",
              "name": "NOC - 3494839",
              "@referredType": "Organisation"
            }
          }
        ],
        "relatedEntity": [
          {
            "@type": "RelatedEntityRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedEntityRefOrValue.schema.json",
            "role": "associated cost",
            "entity": {
              "@type": "EntityRef",
              "@schemaLocation": "http://my.schemas/Entity.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
              "id": "9866",
              "@referredType": "ProjectedCost"
            }
          }
        ],
```


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```
        "note": [
          {
            "@type": "Note",
            "@schemaLocation": "http://my.schemas/Note.schema.json",
            "id": "789789789",
            "author": "NOC operator",
            "date": "2025-04-23T04:16:56Z",
            "text": "Truck roll required"
          }
        ],
        "lastUpdate": "2025-04-23T04:16:56Z",
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
            "owner": "regulatory",
            "externalIdentifierType": "regReport",
            "id": "REGREPORT-124564879"
          }
        ]
      }
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "OutageCreateEvent"
  }
```


Delete Event

Message example for OutageDeleteEvent event


```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "OutageDeleteEvent illustration",
    "domain": "Network Assurance",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2024-08-14T11:03:08.176Z",
    "eventType": "OutageDeleteEvent",
    "priority": "5",
    "timeOccurred": "2024-08-14T11:03:08.176Z",
    "title": "OutageDeleteEvent",
    "event": {
      "outage": {
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
456",
        "id": "123-456",
        "@type":"Outage"
      }
```


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```
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "OutageDeleteEvent"
  }
```

Attribute Value Change Event

Message example for Outage Attribute Value Change Event event

```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "OutageAttributeValueChangeEvent illustration",
    "domain": "Network Assurance",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2025-04-23T04:16:56.176Z",
    "eventType": "OutageAttributeValueChangeEvent",
    "priority": "5",
    "timeOccurred": "2025-04-23T04:16:56.176Z",
    "title": "OutageAttributeValueChangeEvent",
    "event": {
      "outage": {
        "@type": "Outage",
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
456",
        "id": "123-456",
        "estimatedDateToResolve": "2025-04-23T12:16:56Z",
        "estimatedNextUpdate": "2025-04-23T06:16:56Z",
        "lastUpdate": "2025-04-23T04:16:56Z"
      }
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
    "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "OutageAttributeValueChangeEvent"
  }
```

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State Change Event

Message example for OutageStateChangeEvent event

```
Content-Type: application/json
{
    "correlationId": "2cdbfa6e-6756",
    "description": "OutageStateChangeEvent illustration",
    "domain": "Network Assurance",
    "eventId": "41af-bb8d-b056f61b8696",
    "eventTime": "2025-04-23T04:16:56.176Z",
    "eventType": "OutageStateChangeEvent",
    "priority": "5",
    "timeOccurred": "2025-04-23T04:16:56.176Z",
    "title": "OutageStateChangeEvent",
    "event": {
      "outage": {
        "@type": "Outage",
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
456",
        "id": "123-456",
        "state": "partiallyRestored",
        "stateChangeDate": "2025-04-23T04:16:56Z",
        "stateChangeReason" : "restoration in progress remotely",
        "lastUpdate": "2025-04-23T04:16:56Z"
      }
    },
    "reportingSystem": {
        "id": "819",
        "name": "APP-913",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "152",
        "name": "APP-335",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "OutageStateChangeEvent"
  }
```

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Outage

Retrieves a Outage by ID

`GET /outage/{id}?fields=…`

Description

This operation retrieves an Outage entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a spesific Outage with id 3180.

Request

```
GET /outage/3180
Content-Type: application/json
```

Response

```
200
{
```

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```
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/3180",
    "id": "3180",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "initiated",
    "stateChangeDate": "2025-04-23T04:16:56Z",
    "stateChangeReason": "identified from INC-789789",
    "impact": "significant",
    "outageType": "significant local",
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
```


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```
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Truck roll required"
      }
    ],
    "lastUpdate": "2025-04-23T04:16:56Z",
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
```


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```
        "id": "REGREPORT-124564879"
      }
    ]
  }
```

List or find Outage objects

`GET /outage?fields=…`

Description

This operation list Outage entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of Outage(s). The given filter criteria is impact minor and fields id,href,name,@type,impact

Request

```
GET /outage?filter=impact=minor&fields=id,href,name,@type,impact
Content-Type: application/json
```

Response

```
200
[
        {
          "@type": "Outage",
          "href": "http://serverlocation:port/tmf-
api/outageManagement/v5/outage/123-500",
          "id": "123-500",
          "name": "QWR23 FDH down",
          "impact": "minor"
        },
        {
          "@type": "Outage",
          "href": "http://serverlocation:port/tmf-
api/outageManagement/v5/outage/123-501",
          "id": "123-501",
          "name": "QWR24 FDH down",
          "impact": "minor"
        },
        {
          "@type": "Outage",
          "href": "http://serverlocation:port/tmf-
api/outageManagement/v5/outage/123-502",
          "id": "123-502",
          "name": "QWR25 FDH down",
          "impact": "minor"
        }
]
```

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Here's an example of a request for retrieving a list of Outage(s).

Request

```
GET /outage
Content-Type: application/json
```

Response

```
200
[
    {
        "@type": "Outage",
        "@schemaLocation": "http://my.schemas/Outage.schema.json",
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
456",
        "id": "123-456",
        "name": "QWR23 FDH down",
        "description": "broadband outage due to software failure at FDH",
        "isPlanned": false,
        "domain": "broadband",
        "reason": "software failure at FDH",
        "state": "declared",
        "stateChangeDate": "2025-04-23T04:16:56Z",
        "stateChangeReason": "identified from INC-789789",
        "impact": "significant",
        "outageType": "significant local",
        "creationDate": "2025-04-23T04:16:56Z",
        "estimatedDateToResolve": "2025-04-23T10:16:56Z",
        "estimatedNextUpdate": "2025-04-23T05:16:56Z",
        "period": {
          "startDateTime": "2025-04-23T04:16:56Z"
        },
        "rootCauseIncident": [
          {
            "@type": "IncidentRef",
            "@schemaLocation": "http://my.schemas/Incident.schema.json",
            "href": "http://serverlocation:port/tmf-
api/Incident/v4/incident/789789",
            "id": "789789",
            "name": "Incident - 59840 software failure",
            "@referredType": "Incident"
          }
        ],
        "affectedLocation": [
          {
            "@type": "RelatedPlaceRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedPlaceRefOrValue.schema.json",
            "role": "localDistributionPoint",
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
              "id": "2435",
              "name": "FDH location",
              "@referredType":"GeographicLocation"
            }
```

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```
          }
        ],
        "totalNumberOfServiceImpacted": 3,
        "totalNumberOfLocationImpacted": 1,
        "affectedService": [
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
            "id": "123699",
            "name": "broadband connectivity service",
            "@referredType":"Service"
          },
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
            "id": "123700",
            "name": "broadband connectivity service",
            "@referredType":"Service"
          },
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
            "id": "123701",
            "name": "broadband connectivity service",
            "@referredType":"Service"
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
            "role": "restorationCoordinator",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
              "id": "9866",
              "name": "NOC - 3494839",
              "@referredType": "Organisation"
            }
          }
        ],
        "relatedEntity": [
          {
            "@type": "RelatedEntityRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedEntityRefOrValue.schema.json",
            "role": "associated cost",
            "entity": {
              "@type": "EntityRef",
              "@schemaLocation": "http://my.schemas/Entity.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
              "id": "9866",
              "@referredType":"ProjectedCost"
```


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```
            }
          }
        ],
        "note": [
          {
            "@type": "Note",
            "@schemaLocation": "http://my.schemas/Note.schema.json",
            "id": "789789789",
            "author": "NOC operator",
            "date": "2025-04-23T04:16:56Z",
            "text": "Truck roll required"
          }
        ],
        "lastUpdate": "2025-04-23T04:16:56Z",
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
            "owner": "regulatory",
            "externalIdentifierType": "regReport",
            "id": "REGREPORT-124564879"
          }
        ]
    },
    {
        "@type": "Outage",
        "@schemaLocation": "http://my.schemas/Outage.schema.json",
        "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-
457",
        "id": "123-457",
        "name": "QWR24 FDH down",
        "description": "broadband outage due to software failure at FDH",
        "isPlanned": false,
        "domain": "broadband",
        "reason": "software failure at FDH",
        "state": "declared",
        "stateChangeDate": "2025-04-23T04:16:56Z",
        "stateChangeReason": "identified from INC-789789",
        "impact": "minor",
        "outageType": "minor local",
        "creationDate": "2025-04-23T04:16:56Z",
        "estimatedDateToResolve": "2025-04-23T10:16:56Z",
        "estimatedNextUpdate": "2025-04-23T05:16:56Z",
        "period": {
          "startDateTime": "2025-04-23T04:16:56Z"
        },
        "rootCauseIncident": [
          {
            "@type": "IncidentRef",
            "@schemaLocation": "http://my.schemas/Incident.schema.json",
            "href": "http://serverlocation:port/tmf-
api/Incident/v4/incident/456123",
            "id": "456123",
            "name": "Incident - 598989 software failure",
            "@referredType": "Incident"
          }
        ],
        "affectedLocation": [
          {
            "@type": "RelatedPlaceRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedPlaceRefOrValue.schema.json",
            "role": "localDistributionPoint",
```


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```
            "place": {
              "@type": "PlaceRef",
              "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2436",
              "id": "2436",
              "name": "FDH location",
              "@referredType":"GeographicLocation"
            }
          }
        ],
        "totalNumberOfServiceImpacted": 2,
        "totalNumberOfLocationImpacted": 1,
        "affectedService": [
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123800",
            "id": "123800",
            "name": "broadband connectivity service",
            "@referredType":"Service"
          },
          {
            "@type": "ServiceRef",
            "@schemaLocation": "http://my.schemas/Service.schema.json",
            "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123801",
            "id": "123801",
            "name": "broadband connectivity service",
            "@referredType":"Service"
          }
        ],
        "relatedParty": [
          {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
            "role": "restorationCoordinator",
            "partyOrPartyRole": {
              "@type": "PartyRef",
              "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
              "id": "9866",
              "name": "NOC - 3494839",
              "@referredType": "Organisation"
            }
          }
        ],
        "relatedEntity": [
          {
            "@type": "RelatedEntityRefOrValue",
            "@schemaLocation":
"http://my.schemas/RelatedEntityRefOrValue.schema.json",
            "role": "associated cost",
            "entity": {
              "@type": "EntityRef",
              "@schemaLocation": "http://my.schemas/Entity.schema.json",
              "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
              "id": "9866",
              "@referredType":"ProjectedCost"
```


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```
            }
          }
        ],
        "note": [
          {
            "@type": "Note",
            "@schemaLocation": "http://my.schemas/Note.schema.json",
            "id": "789789789",
            "author": "NOC operator",
            "date": "2025-04-23T04:16:56Z",
            "text": "Truck roll required"
          }
        ],
        "lastUpdate": "2025-04-23T04:16:56Z",
        "externalIdentifier": [
          {
            "@type": "ExternalIdentifier",
            "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
            "owner": "regulatory",
            "externalIdentifierType": "regReport",
            "id": "REGREPORT-124564880"
          }
        ]
    }
]
```


Creates a Outage

`POST /outage?fields=…`

Description

This operation creates an Outage entity.

Mandatory Attributes

Mandatory Attributes	Rule
isPlanned	
name	
reason	
@type	

Usage samples

Creation of a new Outage with POST operation

Request

```
POST /outage
Content-Type: application/json
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "name": "QWR23 FDH down",
```

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```
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "initiated",
    "stateChangeReason": "identified from INC-789789",
    "impact": "significant",
    "outageType": "significant local",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
```


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```
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ]
  }
```


Response


```
201
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/123-456",
    "id": "123-456",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "initiated",
    "stateChangeDate": "2025-04-23T04:16:56Z",
    "stateChangeReason": "identified from INC-789789",
    "impact": "significant",
    "outageType": "significant local",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
```


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```
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
```


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```
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Truck roll required"
      }
    ],
    "lastUpdate": "2025-04-23T04:16:56Z",
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
        "id": "REGREPORT-124564879"
      }
    ]
  }
```


Updates partially a Outage

`PATCH /outage/{id}?fields=…`

Description

This operation allows partial updates of an Outage entity. Support of json/merge

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(https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
lastUpdate	
stateChangeDate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
affectedLocation	
affectedService	
description	
domain	
estimatedDateToResolve	
estimatedNextUpdate	
externalIdentifier	
impact	
isPlanned	
name	
note	
outageType	
period	
reason	
relatedEntity	
relatedParty	
restoredLocation	
restoredService	
rootCauseChangeRequest	

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Patchable Attributes	Rule
rootCauseIncident	
state	
stateChangeReason	
totalNumberOfLocationImpacted	
totalNumberOfServiceImpacted	

Usage samples

Here's an example of a request for updating an Outage - set state to declared, update stateChangeReason and estimatedDateToResolve. This example illustrating patch application/json.

Request

```
PATCH /outage/3180
Content-Type: application/json
{
    "state": "declared",
    "stateChangeReason": "Outage is confirmed and must be declared",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "@type": "Outage"
}
```

Response

```
200
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/3180",
    "id": "3180",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "declared",
    "stateChangeDate": "2025-04-23T06:16:56Z",
    "stateChangeReason": "Outage is confirmed and must be declared",
    "impact": "significant",
    "outageType": "significant local",
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
```

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```
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
```


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```
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Truck roll required"
      }
    ],
    "lastUpdate": "2025-04-23T06:16:56Z",
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
        "id": "REGREPORT-124564879"
      }
    ]
  }
```


Here's an example of a request for updating an Outage - set state to declared, update stateChangeReason and estimatedDateToResolve. This example illustrating patch merge, When PATCH is implemeted merge syntax is mandatory

Request

```
PATCH /outage/3180
Content-Type: application/merge-patch+json
{
    "state": "declared",
    "stateChangeReason": "Outage is confirmed and must be declared",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "@type": "Outage"
}
```

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Response


```
200
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/3180",
    "id": "3180",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "declared",
    "stateChangeDate": "2025-04-23T06:16:56Z",
    "stateChangeReason": "Outage is confirmed and must be declared",
    "impact": "significant",
    "outageType": "significant local",
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
```


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```
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Truck roll required"
      }
    ],
    "lastUpdate": "2025-04-23T06:16:56Z",
```


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```
    "externalIdentifier": [
      {
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
        "id": "REGREPORT-124564879"
      }
    ]
  }
```

Here's an example of a request for updating an Outage - set state to declared, update stateChangeReason, estimatedDateToResolve and add new Note. This example illustrating patch json

Request

```
PATCH /outage/3180
Content-Type: application/json-patch+json
[
    {
        "op":"replace",
        "path": "/state",
        "value": "declared"
    },
    {
        "op":"replace",
        "path": "/stateChangeReason",
        "value": "Outage is confirmed and must be declared"
    },
    {
        "op":"replace",
        "path": "/estimatedDateToResolve",
        "value": "2025-04-23T10:16:56Z"
    },
    {
        "op": "add",
        "path":"/note",
        "value":
        {
            "@type": "Note",
            "id": "77456",
            "author": "Jack Smith",
            "date": "2025-04-23T07:16:56Z",
            "text": "This is quite important, please approach me as soon as
possible"
        }
    }
  ]
```

Response

```
200
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/3180",
```

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```
    "id": "3180",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned": false,
    "domain": "broadband",
    "reason": "software failure at FDH",
    "state": "declared",
    "stateChangeDate": "2025-04-23T04:16:56Z",
    "stateChangeReason": "Outage is confirmed and must be declared",
    "impact": "minor",
    "outageType": "significant local",
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
```


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```
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123701",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Truck roll required"
      },
      {
        "@type": "Note",
        "id": "77456",
        "author": "Jack Smith",
        "date": "2025-04-23T07:16:56Z",
        "text": "This is quite important, please approach me as soon as possible"
      }
    ],
    "lastUpdate": "2025-04-23T06:20:56Z",
    "externalIdentifier": [
      {
```


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```
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
        "id": "REGREPORT-124564879"
      }
    ]
  }
```

Here's an example of a request for updating an Outage - set state to partiallyRestored, update stateChangeReason, note, affectedService and restoredService. This example illustrating patch query json

Request

```
PATCH /outage/3180
Content-Type: application/json-patch-query+json
[
    {
        "op":"replace",
        "path": "/state",
        "value": "partiallyRestored"
    },
    {
        "op":"replace",
        "path": "/stateChangeReason",
        "value": "restoration in progress remotely"
    },
    {
        "op": "replace",
        "path":"/note?id=789789789",
        "value":
        {
            "date": "2025-04-23T04:20:56Z",
            "text": "Restoration expedited remotely"
        }
    },
    {
        "op": "remove",
        "path":"/affectedService?id=123699"
    },
    {
        "op": "remove",
        "path":"/affectedService?id=123700"
    },
    {
        "op": "add",
        "path":"/restoredService?id=123699"
    },
    {
        "op": "add",
        "path":"/restoredService?id=123700"
    }
  ]
```

Response

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```
200
{
    "@type": "Outage",
    "@schemaLocation": "http://my.schemas/Outage.schema.json",
    "href": "http://serverlocation:port/tmf-api/outageManagement/v5/outage/3180",
    "id": "3180",
    "name": "QWR23 FDH down",
    "description": "broadband outage due to software failure at FDH",
    "isPlanned":false,
    "domain": "broadband",
    "reason": "restoration in progress remotely",
    "state": "partiallyRestored",
    "stateChangeDate": "2025-04-23T04:16:56Z",
    "stateChangeReason": "restoration in progress remotely",
    "impact": "minor",
    "outageType": "significant local",
    "creationDate": "2025-04-23T04:16:56Z",
    "estimatedDateToResolve": "2025-04-23T10:16:56Z",
    "estimatedNextUpdate": "2025-04-23T05:16:56Z",
    "period": {
      "startDateTime": "2025-04-23T04:16:56Z"
    },
    "rootCauseIncident": [
      {
        "@type": "IncidentRef",
        "@schemaLocation": "http://my.schemas/Incident.schema.json",
        "href": "http://serverlocation:port/tmf-api/Incident/v4/incident/789789",
        "id": "789789",
        "name": "Incident - 59840 software failure",
        "@referredType": "Incident"
      }
    ],
    "affectedLocation": [
      {
        "@type": "RelatedPlaceRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedPlaceRefOrValue.schema.json",
        "role": "localDistributionPoint",
        "place": {
          "@type": "PlaceRef",
          "@schemaLocation": "http://my.schemas/PlaceRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/geographicLocationManagement/v5/geographicLocation/2435",
          "id": "2435",
          "name": "FDH location",
          "@referredType":"GeographicLocation"
        }
      }
    ],
    "totalNumberOfServiceImpacted": 3,
    "totalNumberOfLocationImpacted": 1,
    "affectedService": [
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123701",
        "id": "123710",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "restoredService": [
```

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```
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123699",
        "id": "123699",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      },
      {
        "@type": "ServiceRef",
        "@schemaLocation": "http://my.schemas/Service.schema.json",
        "href": "http://serverlocation:port/tmf-
api/serviceInventory/v5/service/123700",
        "id": "123700",
        "name": "broadband connectivity service",
        "@referredType":"Service"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation":
"http://my.schemas/RelatedPartyRefOrPartyRoleRef.schema.json",
        "role": "restorationCoordinator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "http://my.schemas/PartyRef.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/organisation/9866",
          "id": "9866",
          "name": "NOC - 3494839",
          "@referredType": "Organisation"
        }
      }
    ],
    "relatedEntity": [
      {
        "@type": "RelatedEntityRefOrValue",
        "@schemaLocation": "http://my.schemas/RelatedEntityRefOrValue.schema.json",
        "role": "associated cost",
        "entity": {
          "@type": "EntityRef",
          "@schemaLocation": "http://my.schemas/Entity.schema.json",
          "href": "https://mycsp.com:8080/tmf-
api/costManagement/v5/projectedCost/9866",
          "id": "9866",
          "@referredType":"ProjectedCost"
        }
      }
    ],
    "note": [
      {
        "@type": "Note",
        "@schemaLocation": "http://my.schemas/Note.schema.json",
        "id": "789789789",
        "author": "NOC operator",
        "date": "2025-04-23T04:16:56Z",
        "text": "Restoration expedited remotely"
      }
    ],
    "lastUpdate": "2025-04-23T04:20:56Z",
    "externalIdentifier": [
```


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```
      {
        "@type": "ExternalIdentifier",
        "@schemaLocation": "http://my.schemas/ExternalIdentifier.schema.json",
        "owner": "regulatory",
        "externalIdentifierType": "regReport",
        "id": "REGREPORT-124564879"
      }
    ]
  }
```

Deletes a Outage

`DELETE /outage/{id}`

Description

This operation deletes an Outage entity.

Usage samples

Here's an example of a request for delete a specific Outage.

Request

```
DELETE /outage/3180
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Version 5.0.0	June 2025	Anu Aulakh Telstra anu.aulakh@team.telstra.com	First release
Version 5.0.0	June 2025	Varun Sasidharan Nair Telstra varun.nair@team.telstra.com	First release
Version 5.0.0	June 2025	Abdul Majid Hussain Telstra abdul.majidhussain@team.telstra.com	First release

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